

G-CAM V6.18

Cloudflare Usage Requirements & Free vs Paid Recommendation

Script-specific deployment report for install_v6.18.sh

EXECUTIVE VERDICT

For the current G-Cam architecture, Cloudflare Free is technically sufficient for the permanent named Cloudflare Tunnel, DNS, HTTPS, CDN/DDoS baseline and basic WAF. Paying for Pro is not required to make the tunnel permanent. For a real deployed camera system, the recommended baseline is Free + a named Tunnel + Cloudflare Access protection for sensitive web paths. Upgrade to Pro for stronger WAF/bot controls and support; upgrade to Business when a contractual uptime SLA or higher-tier support is required.

Scope

Permanent dashboard, live MJPEG view, playback, camera configuration proxy, Cloudflare Tunnel service, watchdog and temporary TryCloudflare access.

Based on the uploaded G-Cam V6.18 installer and current Cloudflare documentation.

1. How G-Cam Uses Cloudflare

The installer is not merely using Cloudflare DNS. It provisions a persistent connector on the Raspberry Pi and publishes the local Flask application through a named Cloudflare Tunnel.

Stage	G-Cam implementation	Purpose
1. Camera / recordings	SUB stream for AI + public live; MAIN stream for HQ SFTP event media	Separates lightweight remote viewing from high-quality capture
2. Local G-Cam app	Flask service listens on port 8080	Dashboard, APIs, live view, playback and camera proxy
3. cloudflared	Persistent systemd service, token file, metrics on 127.0.0.1:20241	Outbound-only secure connector to Cloudflare
4. Cloudflare hostname	Named public hostname routes to http://localhost:8080	Stable HTTPS URL without opening inbound router ports
5. Watchdog	Checks local origin, cloudflared readiness and public URL every minute	Restarts G-Cam or tunnel after repeated failures

REMOTE USER -> CLOUDFLARE EDGE -> NAMED TUNNEL -> cloudflared ON PI -> FLASK :8080

DNS / HTTPS / WAF / Access outbound TCP/UDP 7844 /live /playback /camera

Script-specific performance profile

The public Cloudflare live stream is deliberately conservative: 640-pixel width, JPEG quality 45 and target 3 FPS. That helps reduce Raspberry Pi CPU, network load and browser buffering. This is much lighter than the local dashboard live settings.

- Permanent URLs exposed by the script include the dashboard root, /live, /playback and /camera.
- The installer stores the named-tunnel token in a root-owned file with mode 600 and runs cloudflared with --token-file.
- The watchdog uses the connector /ready endpoint plus local and public HTTP checks.
- The script also contains a temporary TryCloudflare flow. Cloudflare documents Quick Tunnels as testing/development only, so it should remain temporary rather than replace the named production tunnel.

2. Exact Cloudflare Requirements for This Script

Cloudflare-side requirements

Requirement	Needed?	Notes
Cloudflare account	YES	Required to create and manage the named tunnel.
Domain active on Cloudflare	YES	The production hostname must be in an active Cloudflare zone.
Named Cloudflare Tunnel	YES	This is the permanent production route expected by the installer.
Published application route	YES	Map <device-subdomain>.<your-domain> to http://localhost:8080.
Tunnel token	YES	Installer expects a token copied from the tunnel dashboard.
Paid Web plan (Pro)	NO	Tunnel is available on all Cloudflare plans.
Paid Access seats	NO for <=50 users	Cloudflare One Free currently supports up to 50 users; Access is a separate product from the Website Pro plan.
Public IPv4 / port-forwarding	NO	Tunnel is outbound-only; no inbound public origin port is required.

Network and Raspberry Pi requirements

- Local G-Cam application must be reachable at http://127.0.0.1:8080 / http://localhost:8080.
- Outbound TCP and UDP port 7844 must be permitted for Cloudflare Tunnel (HTTP/2 uses TCP; QUIC uses UDP).
- Working DNS and general Internet connectivity are required.
- The installer supports arm64 and armhf cloudflared packages and rejects an unsupported architecture.
- The installer requires cloudflared 2025.4.0 or newer because it uses secure --token-file support.
- Protocol "auto" is the sensible default unless a firewall/network requires forcing http2 or quic.

No router port-forwarding is needed

Cloudflare Tunnel creates outbound connections from the Pi to Cloudflare. This is one of the strongest reasons to keep the named Tunnel architecture: the G-Cam origin does not need a publicly routable IP or an inbound port opened on the customer router.

3. Free vs Paid: What Actually Changes for G-Cam

The website plan affects the depth of Cloudflare security, performance features and support. It does not determine whether the named Tunnel exists or is permanent.

Item	Free	Pro	Business	Impact on G-Cam
Named Cloudflare Tunnel	Included	Included	Included	Free is enough for the tunnel itself.
DNS + Universal SSL	Included	Included	Included	Stable HTTPS hostname works on Free.
Unmetered DDoS + CDN baseline	Included	Included	Included	Core edge protection exists on Free.
WAF	Basic / Free managed rules	Deeper managed rules + stronger controls	Higher controls	Pro is useful if the public G-Cam hostname sees hostile Internet traffic.
Bot protection	Basic	Stronger	Stronger still	Helpful, but not required for normal Tunnel operation.
Technical support	Community-focused	Ticket support	Ticket + chat	Paid plan matters when support response is operationally important.
Uptime SLA	No	No	100% SLA	Business is the first self-serve plan with a published 100% SLA.
Price (current)	US\$0/mo	US\$20/mo annual or US\$25 monthly	US\$200/mo annual or US\$250 monthly	Web plan price is per Cloudflare zone/domain, separate from Zero Trust seats.

Best value for the current G-Cam design

Start with Cloudflare Free for the domain + named Tunnel. Add Cloudflare Access on sensitive paths. Move to Pro when you want stronger WAF/bot controls or ticket support. Do not buy Pro only because you need a permanent tunnel - that capability is already available on Free.

Cloudflare Access is a separate decision

Cloudflare Access belongs to Cloudflare One / Zero Trust, not to the Free-vs-Pro website plan. The current Cloudflare One Free tier supports up to 50 users. If a deployment needs more seats, Cloudflare currently lists a pay-as-you-go tier at US\$7 per user/month. This means a G-Cam can use a Free website plan and still have identity-based protection for a small authorized user group.

4. Security Findings and Recommended Access Policy

HIGH PRIORITY - protect the camera configuration proxy

The script exposes /camera as a general proxy to the camera configuration interface and supports GET, POST, PUT, PATCH, DELETE and OPTIONS. The shown route handler does not perform an application-level login check before proxying the request. Do not leave this path openly accessible on an Internet-facing hostname.

Path / function	Current role	Recommended Cloudflare control
/camera and /camera/*	Camera configuration proxy	Cloudflare Access: allow only administrators.
/playback and /playback/*	Stored DVR playback	Cloudflare Access: allow authorized operators/viewers.
Dashboard root + control APIs	Status and controls	Cloudflare Access: authenticate users; keep control tokens as a second layer, not the only layer.
/live and video feed	Remote camera live view	Protect with Access unless the live view is intentionally public.
/api/state	Public watchdog health target	Use a service-authenticated health check or a deliberately minimal health endpoint.

Watchdog + Access interaction

The current public watchdog treats any 2xx, 3xx or 4xx response as "healthy." If Cloudflare Access is placed in front of the whole hostname, an unauthenticated request can receive an Access redirect or denial before the origin is reached, and the watchdog may still count that response as healthy. For a hardened deployment, modify the watchdog to authenticate with an Access service token and require a true origin health response, or use a separate minimal health route designed for monitoring.

- Keep the tunnel token root-owned and mode 600 as the installer already does.
- Do not reuse the temporary TryCloudflare URL as a production endpoint. Cloudflare explicitly positions Quick Tunnels for testing/development and does not provide an SLA for them.
- Review and rotate credentials embedded in deployment scripts. Move long-lived secrets into protected configuration/secret storage rather than source code whenever practical.
- Do not cache MJPEG/live responses or control/API responses. The script already sets no-cache/no-store for several live/playback responses; preserve that behavior when adding Cloudflare cache rules.
- Use WAF/rate limiting around control and login-like endpoints if the hostname is exposed to the public Internet.

5. Recommended Production Configuration

Recommended baseline

Cloudflare Website Plan: FREE | Tunnel: NAMED / REMOTELY MANAGED | Access: FREE (up to 50 users) | Protocol: AUTO | Origin: http://localhost:8080

Cloudflare dashboard steps

1. Add the production domain to Cloudflare and ensure the zone is active.
2. Go to Networking > Tunnels and create a named tunnel for the G-Cam device/site.
3. Add a Published application route for <device-subdomain>.<your-domain>.
4. Set Service to http://localhost:8080.
5. Copy the tunnel token / Add replica command and supply the token to the G-Cam installer.
6. Keep transport protocol set to auto unless network policy requires http2 or quic.
7. Confirm the network allows outbound TCP/UDP port 7844.
8. Create Cloudflare Access applications/policies for /camera, playback, dashboard/control paths and live view as appropriate.
9. For the public health check, implement an Access service token or a minimal monitoring endpoint before enforcing Access over the entire hostname.

Post-install verification

Check	Expected result
systemctl status cloudflared.service	active (running)
Cloudflared readiness endpoint	HTTP 200 on 127.0.0.1:20241/ready
Local G-Cam health	Local :8080 health/API responds
Permanent HTTPS hostname	Responds through the named tunnel
Watchdog timer	Runs approximately every 60 seconds
/live	Remote view loads at the intended public quality
/playback	Authorized user can retrieve stored recordings
/camera	Blocked to unauthenticated Internet users; works for authorized administrators

6. When Should You Upgrade?

Deployment situation	Recommended plan	Why
Lab / development / pilot	Free	Named Tunnel, DNS/SSL and baseline security are sufficient.
Small production G-Cam fleet, <=50 authorized operators	Free + Access Free	Best cost/security balance if operational SLA is not contractual.
Internet-exposed commercial deployment needing stronger WAF/bot controls or ticket support	Pro + Access	Upgrade for security depth/support, not for Tunnel permanence.
Business-critical deployment requiring contractual uptime and chat support	Business + Access	Published 100% uptime SLA and higher support tier.
Large fleet, regulated environment, custom security/SLA	Contract / Enterprise	Custom support, controls, architecture and commercial terms.

For your current script: FREE is sufficient, but "Free and public" is not the same as "secure."

The core technical requirement is a named Cloudflare Tunnel and an active Cloudflare-managed domain. The most valuable next control is identity/access protection on the sensitive G-Cam web paths. Pro becomes justified when the deployment needs stronger web security controls and support. Business becomes justified when an SLA is a business requirement.

Capacity considerations

- Each remote MJPEG viewer creates continuous traffic. The current 640px / quality 45 / 3 FPS public stream is intentionally constrained, but real bandwidth still depends on scene complexity and JPEG size.
- For many simultaneous viewers or a large device fleet, measure Cloudflare and origin bandwidth/CPU rather than assuming the website plan alone will solve capacity.
- Cloudflare Tunnel can run additional replicas for availability; a second connector host provides origin-side redundancy, but the current installer provisions one Pi connector per device.
- If video distribution becomes the primary workload at scale, evaluate a video-specific delivery architecture rather than relying indefinitely on a direct MJPEG stream from each Pi.

7. Source Notes

Script-derived observations are based on the user-provided `install_v6.18.sh` (G-Cam V6.18). Sensitive credentials present in the installer are intentionally not reproduced in this report.

S1 G-Cam V6.18 installer: Cloudflare hostname, named-tunnel token, local service mapping, cloudflared systemd unit, watchdog and public live settings. User-provided `install_v6.18.sh`.

C1 Cloudflare Tunnel documentation - "Available on all plans" and outbound-only origin connectivity.
<https://developers.cloudflare.com/tunnel/>

C2 Cloudflare Published applications - public hostname routes to local services; paid Access is not required merely to publish via Tunnel. <https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/routing-to-tunnel/>

C3 Cloudflare protocols for published applications - HTTP public hostname can proxy to `http://localhost:<port>`.
<https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/routing-to-tunnel/protocols/>

C4 Cloudflare Tunnel with firewall - outbound port 7844 for TCP/HTTP2 and UDP/QUIC.
<https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/configure-tunnels/tunnel-with-firewall/>

C5 Cloudflare Quick Tunnels - development/testing only; no SLA; 200 in-flight request limit for Quick Tunnels.
<https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/do-more-with-tunnels/trycloudflare/>

C6 Cloudflare plan pricing and features - Free \$0; Pro \$20/mo annual or \$25 monthly; Business \$200/mo annual or \$250 monthly; Business 100% uptime SLA. <https://www.cloudflare.com/plans/>

C7 Cloudflare One pricing - Free for teams under 50 users; pay-as-you-go currently \$7/user/month.
<https://www.cloudflare.com/plans/>

C8 Cloudflare Access self-hosted applications - deny-by-default access policies for protected apps.
<https://developers.cloudflare.com/cloudflare-one/access-controls/applications/http-apps/self-hosted-public-app/>

C9 Cloudflare Access service tokens - machine-to-machine authentication headers for protected applications.
<https://developers.cloudflare.com/cloudflare-one/access-controls/service-credentials/service-tokens/>

Final recommendation

Deploy the current G-Cam script with Cloudflare Free + a permanent named Tunnel. Add Cloudflare Access before exposing camera configuration, playback, controls or private live views. Upgrade to Pro only when stronger WAF/bot protection or ticket support is required; choose Business when a contractual uptime SLA is required.

Pricing and Cloudflare product limits can change. Values in this report were verified against Cloudflare sources on 24 Aug 2026.